SULZER

Original instructions

Installation, operation and maintenance instructions

Submersible Drainage Pump GWP

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1. Important notice

	NOTE
!	The original version of this document is in English. All other languages are a translation of the original. In case of a discrepancy, the English version will prevail.

	NOTE
!	The layout and wording of the online version of this manual may vary from the printed version. The same information is provided in both.

2. Symbols and notices

	DANGER
	Presence of dangerous voltage

	DANGER
	Danger of an explosion occurring.

	WARNING
	Hot surface - danger of burn or injury.

	WARNING
	Hot liquid - danger of burn or injury.

	CAUTION
	Non-compliance may result in personal injury.

	ATTENTION
!	Non-observance may result in damage to the unit or negatively affect its performance.

	NOTE
!	Important information for particular attention.

3. General

!	NOTE
	Sulzer reserves the right to alter specifications due to technical developments.

3.1. Intended use and application

GWP pumps have been designed for the efficient and reliable pumping of grey water and lightly polluted liquids and can be installed for fixed or transportable applications. They are suitable for modern drainage and water handling systems and are designed for pumping of the following liquids:

- Grey water and lightly polluted liquids.
- Water containing suspended solids.
- Water containing solids and fibrous material.
- Municipal, construction and industrial areas.
- Surface and clean water.

These units must not be used in certain applications e.g. operating within flammable, combustible, chemical, corrosive, or explosive liquids.

!	ATTENTION
	The maximum allowable temperature of the medium pumped is 40 °C / 104 °F.

!	ATTENTION
	Leakage of lubricants could result in pollution of the medium being pumped.

!	ATTENTION
	Always consult with your local Sulzer representative for advice on approved use and application before installing the pump.

3.2. Identification code

Table 1.

e.g. GWP 05 01 - 4.5 / 2 EC - 10	
Hydraulics:	**Motor:**
GWP = Product range **05** = Discharge outlet DN (cm) **01** = Hydraulic number	**4.5** = Motor power P_2 kW x 10 **2** = Number of poles **EC** = Cast iron (CR = Stainless) **10** = Cable length (m)

GWP = Standard pump for efficient pumping.

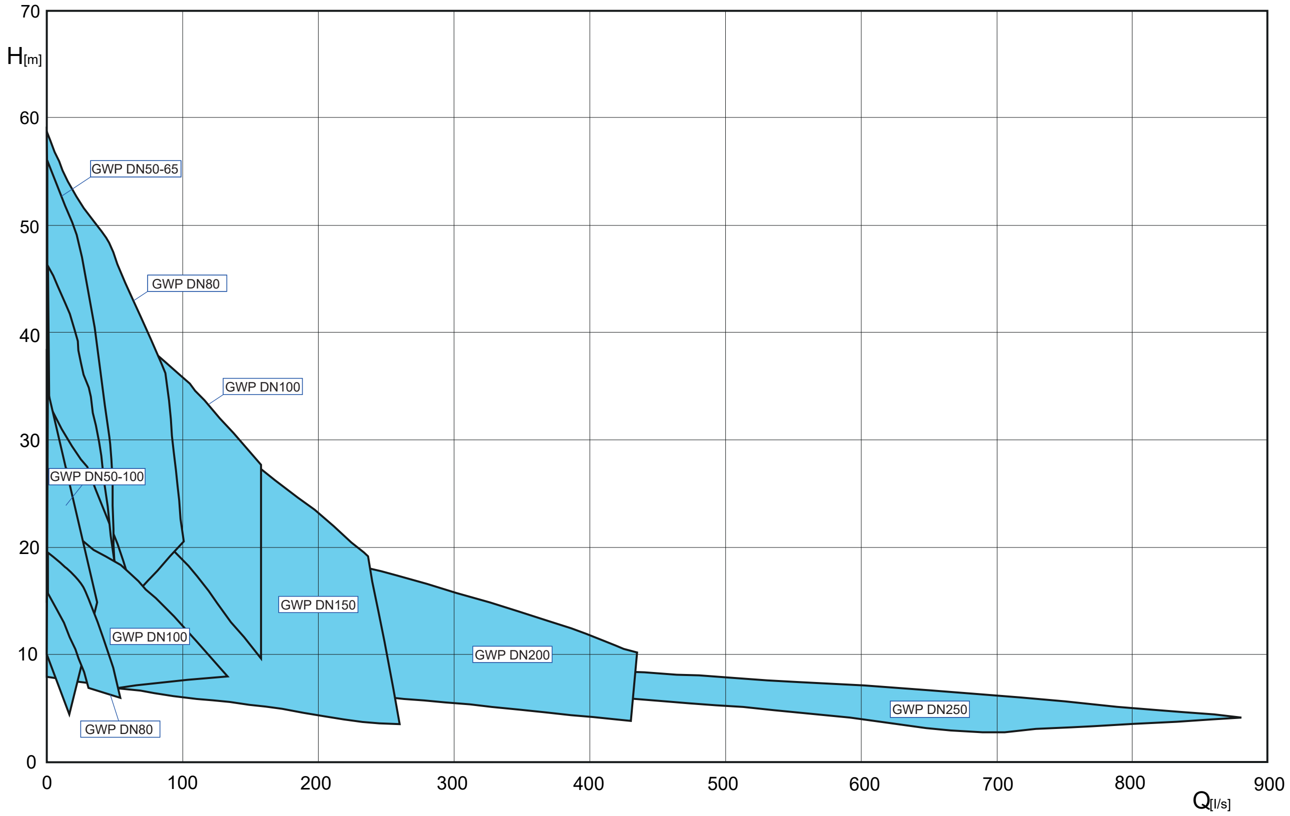
GWPS = Screened pump for applications where larger solids are expected.

GWPC = Cutter pump for solids handling where fibrous materials are present.

Cast Iron (EC) = Rugged.

Stainless steel = Provides superior corrosion resistance.

4. Performance range



5. Safety

The general and specific health and safety guidelines are described in detail in the “Sulzer Safety Instructions” booklet. If anything is unclear, or you have any questions regarding safety make certain to contact the manufacturer Sulzer.

This unit can be used by children aged 8 years and above, and persons with reduced physical, sensory, or mental capabilities, or lack of experience and knowledge, when they have been given supervision or instruction concerning the safe use of the device and understand the hazard involved. Children must not play with the appliance. Cleaning and user maintenance should not be performed by children without supervision.

	CAUTION
	Under no circumstances place a hand inside the suction or discharge openings unless the pump is completely isolated from the power supply.

5.1. Personal protective equipment

Submersible electrical units can present mechanical, electrical, and biological hazards to personnel during installation, operation, and service. It is obligatory that appropriate personal protective equipment (PPE) is used. The minimum requirement is the wearing of safety glasses, footwear, and gloves. However, an on-site risk assessment should always be carried out to determine if additional equipment is required e.g. safety harness, breathing equipment etc.

6. Technical data

Maximum noise level ≤ 70 dB. In some types of installations it is possible that during operation the noise level of 70 dB(A) or the measured noise level may be exceeded.

Detailed technical information is available in the technical data sheet which can be downloaded from https://www.sulzer.com

6.1. Nameplates

We recommend that you record the data from the standard nameplate on the unit in the legend below, and maintain it as a source of reference for the ordering of spare parts, repeat orders and general queries.

Always state the type, item number and serial number in all communications.

6.1.1. Nameplate drawing

Figure 1. Nameplate drawing

Table 2. legend

Legend	Description	Unit
Date	Production date	
Type	Pump type	
Max.	Maximum submergence depth	m / ft
P_N	Power	kW
U_N	Rated voltage	V
I_N	Rated current	A
Cos φ	Power factor	pf

table continued

Legend	Description	Unit
	Address	
$T_A$max.	Ambient temperature	
IP68	Protection method	
Hz	Frequency	Hz
SN	Serial number	
IE	Motor efficiency standard	
H_{max}	Maximum head	m / ft
n / rpm	Speed	r/min / RPM
Weight / Wt	Weight	kg / lbs

6.2. Technical parameter

<table>
<tr><th>Pump model</th><th>Pump designation</th><th>Current 400V (A)</th><th>Weight (kg)</th><th>Cable specifications</th><th>Temperature monitoring</th><th>Leak monitoring</th><th>Discharge (mm)</th><th>Start</th></tr>
<tr><td rowspan="2">GWP</td><td>0501</td><td>1.2</td><td>25</td><td rowspan="6">1x4G1</td><td rowspan="6">Built-in thermal protector</td><td rowspan="6">None</td><td rowspan="8">50</td><td rowspan="14">DOL</td></tr>
<tr><td>0502</td><td>1.9</td><td>26</td></tr>
<tr><td rowspan="12">GWP/ GWPS</td><td>0503</td><td>3.4</td><td>37</td></tr>
<tr><td>0504-0506</td><td>4.8</td><td>40</td></tr>
<tr><td>0507</td><td>6.4</td><td>62</td></tr>
<tr><td>0508</td><td>8.4</td><td>63</td></tr>
<tr><td>0509</td><td>11.1</td><td>81</td><td rowspan="2">1x4G1.5+1x3G1</td><td rowspan="2">PTC(K1/K2)</td><td rowspan="2">Seal chamber/ Motor chamber (K3)</td></tr>
<tr><td>0510-0511</td><td>14.9</td><td>83</td></tr>
<tr><td>0601</td><td>6.4</td><td>63</td><td rowspan="2">1x4G1</td><td rowspan="2">Built-in thermal protector</td><td rowspan="2">None</td><td rowspan="4">65</td></tr>
<tr><td>0602</td><td>8.4</td><td>64</td></tr>
<tr><td>0603</td><td>11.1</td><td>82</td><td rowspan="2">1x4G1.5+1x3G1</td><td rowspan="2">PTC(K1/K2)</td><td rowspan="2">Seal chamber/ Motor Chamber (K3)</td></tr>
<tr><td>0604-0605</td><td>14.9</td><td>84</td></tr>
<tr><td>0801-0803</td><td>3.4</td><td>38</td><td rowspan="2">1x4G1</td><td rowspan="2">Built-in thermal protector</td><td rowspan="2">None</td><td rowspan="2">80</td></tr>
<tr><td>0804-0805</td><td>4.8</td><td>40</td></tr>
</table>

table continued

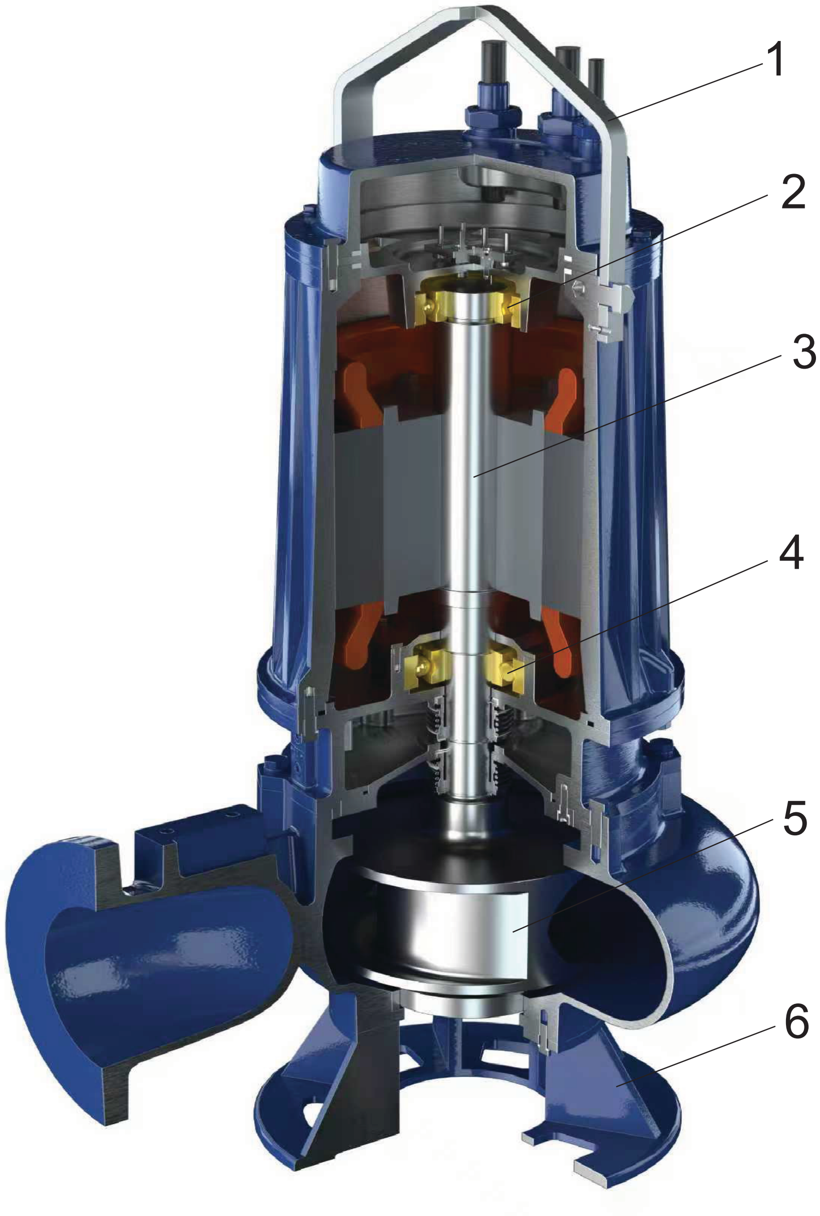
<table>
<tr><th>Pump model</th><th>Pump designation</th><th>Current 400V (A)</th><th>Weight (kg)</th><th>Cable specifications</th><th>Temperature monitoring</th><th>Leak monitoring</th><th>Discharge (mm)</th><th>Start</th></tr>
<tr><td rowspan="6">GWP/
GWPS</td><td>0806-0807</td><td>11.1</td><td>83.5</td><td rowspan="2">1x4G1.5+1x3G1</td><td rowspan="4">PTC(K1/K2)</td><td rowspan="4">Seal chamber/ motor chamber (K3)</td><td rowspan="4">80</td><td rowspan="14">DOL</td></tr>
<tr><td>0808-0809</td><td>14.9</td><td>89.5</td></tr>
<tr><td>0810-0812</td><td>21.8</td><td>144</td><td>1x4G2.5+1x3G1</td></tr>
<tr><td>0813-0816</td><td>29.3</td><td>155</td><td>1x4G4 + 1x3G1</td></tr>
<tr><td>1001</td><td>3.4</td><td>40.5</td><td rowspan="4">1x4G1</td><td rowspan="4">Built-in thermal protector</td><td rowspan="4">None</td><td rowspan="10">100</td></tr>
<tr><td>1002</td><td>4.8</td><td>42</td></tr>
<tr><td rowspan="4">GWP</td><td>1003</td><td>6.4</td><td>66</td></tr>
<tr><td>1004-1005</td><td>8.4</td><td>68</td></tr>
<tr><td>1006</td><td>11.1</td><td>88</td><td rowspan="6">1x4G1.5+1x3G1</td><td rowspan="6">PTC(K1/K2)</td><td rowspan="6">Seal chamber/ motor chamber (K3)</td></tr>
<tr><td>1007</td><td>14.9</td><td>94</td></tr>
<tr><td rowspan="4">GWP/
GWPS</td><td>1008</td><td>11.7</td><td>147</td></tr>
<tr><td>1009</td><td>15.5</td><td>157</td></tr>
<tr><td>1010-1011</td><td>11.7</td><td>147</td></tr>
<tr><td>1012-75/4</td><td>15.5</td><td>157</td></tr>
</table>

table continued

Pump model	Pump designation	Current 400V (A)	Weight (kg)	Cable specifications	Temperature monitoring	Leak monitoring	Discharge (mm)	Start
GWP/ GWPS	1013-1014	22.6	212	1x4G2.5+1x3G1	PTC(K1/K2)	Seal chamber/ motor chamber (K3)	100	DOL
	1015	30.1	223	1x4G4 + 1x3G1				
GWP	1016	30.1	238					
	1017-1018	36.8	268					
	1501	9.1	110				150	
	1502	11.7	116	1x4G1.5+1x3G1				
	1503		160					
	1504	15.5	170					
	1505	22.6	235	1x4G2.5 + 1x3G1				
	1506		238					
	1507	30.1	249	1x4G4 + 1x3G1				
	1508	36.8	272	1x4G6 + 1x3G1				
	1509	42.7	282					
	2001	17.1	182	1x4G1.5+1x3G1			200	
	2002	22.6	261	1x4G2.5 + 1x3G1				
	2003	30.1	272	1x4G4 + 1x3G1				
	2004	36.8	298	1x4G6 + 1x3G1				
	2005	42.8	308					
	2501	24.8	313	1x4G2.5 + 1x3G1			250	
	2502	32.2	335	1x4G4 + 1x3G1				
	2503	38.3	395					
	2504	45	410					
GWPC	0501	3.4	38	1x4G1	Built-in thermal protector	None	50	DOL
	0502	4.8	41					
	0503	6.4	62					
	0504	8.4	66					
	0505							
	1001		71				100	
	1002	11.1	88	1x4G1.5+1x3G1	PTC(K1/K2)	Seal chamber/ motor chamber (K3)		
	1003	14.9	93					

7. General design features

GWP is a submersible grey water pump designed for the efficient and reliable transport of lightly polluted liquids. The water-pressure-tight, encapsulated motor and the hydraulic section form a compact, robust and modular construction suitable for continuous submerged operation.



1 Lifting hoop.
2 Upper bearing.
3 Rotorshaft.
4 Lower bearing.
5 Impeller.
6 Water inlet base.

8. Lifting, transport and storage

8.1. Lifting

!	ATTENTION
	Observe the total weight of the Sulzer units and their attached components! (see nameplate for weight of base unit).

The duplicate nameplate provided must always be located and visible close to where the unit is installed (e.g. at the terminal boxes / control panel where the cables are connected).

!	NOTE
	Lifting equipment must be used if the total unit weight and attached accessories exceeds local manual lifting safety regulations.

The total weight of the unit and accessories must be observed when specifying the safe working load of any lifting equipment! The lifting equipment, e.g. crane and chains, must have adequate lifting capacity. The hoist must be adequately dimensioned for the total weight of the Sulzer units (including lifting chains or steel ropes,and all accessories which may be attached). The end user assumes sole responsibility that lifting equipment is certified, in good condition, and inspected regularly by a competent person at intervals in accordance with local regulations. Worn or damaged lifting equipment must not be used and must be properly disposed of. Lifting equipment must also comply with the local safety rules and regulations

!	NOTE
	The guidelines for the safe use of chains, ropes and shackles supplied by Sulzer are outlined in the Lifting Equipment manual provided with the items and must be fully adhered to.

⚠	⚠ WARNING
	Do not use the power cable to lift the unit, as this may cause damage or injury.

Related information

Nameplate drawing on page 6

8.2. Transport

During transport, care should be taken that the pump cannot fall over or roll and cause damage to the pump or injury to the person. The pumps have a lifting hoop for lifting or suspension of the pump.

⚠	⚠ CAUTION
	After removal from its original packaging we recommend that during future transportation of the pump it is laid on its side and securely strapped to a pallet.

DANGER

Dangerous voltage

The pump must be raised only by the lifting hoop and never by the power cable.

CAUTION

Ensure that the pump is placed on a firm level surface during transport to prevent movement or tipping.

CAUTION

Protect the pump from mechanical damage. Avoid impacts to the hydraulic section and cable entry.

8.3. Storage

ATTENTION

The Sulzer products must be protected from weather influences such as UV from direct sunlight, high humidity, aggressive dust emissions, mechanical damage, frost etc. The Sulzer original packaging with the relevant transport securing devices (where used) ensures optimum protection of the unit. If the units are exposed to temperatures under 0 °C / 32 °F check that there is no water in the hydraulics, cooling system, or other spaces. In the case of heavy frosts, the units and cable should not be moved if possible. When storing under extreme conditions, e.g. in tropical or desert conditions suitable additional protective steps should be taken. We would be glad to advice you further.

NOTE

The Sulzer units normally require no maintenance during storage. During longer storage times, (after approx. one year) the transport locks on the motor shaft (not all versions) must be dismantled. New lubricating oil is applied to the sealing surfaces by manually turning the shaft several times (also for the purpose of cooling or lubricating so that trouble-free function of the sliding ring seal is ensured). No maintenance is required when storing the motor shaft.

8.3.1. Moisture protection of motor connection cable

ATTENTION

The ends of the cables should never be immersed in water as the protective covers only provide protection against water spray or similar (IP44) and are not a water tight seal. The covers should only be removed immediately prior to connecting the pumps electrically.

During storage or installation, prior to the laying and connection of the power cable, particular attention should be given to the prevention of water damage in locations which could flood.

ATTENTION

If there is a possibility of water ingress then the cable should be secured so that the end is above the maximum possible flood level. Take care not to damage the cable or its insulation when doing this.

9. Setup and installation

	CAUTION
	All safety hints in other sections must be observed!

	DANGER
	Dangerous voltage The regulations covering the use of pumps in applications, together with all regulations involving the use of explosion proof motors, should be observed. The cable ducting to the control panel should be sealed off in a gas-tight manner by the use of a foaming material after the cable and control circuits have been pulled through. In particular the safety regulations covering work in enclosed areas in plants should be observed together with general good technical practice.

9.1. Discharge line

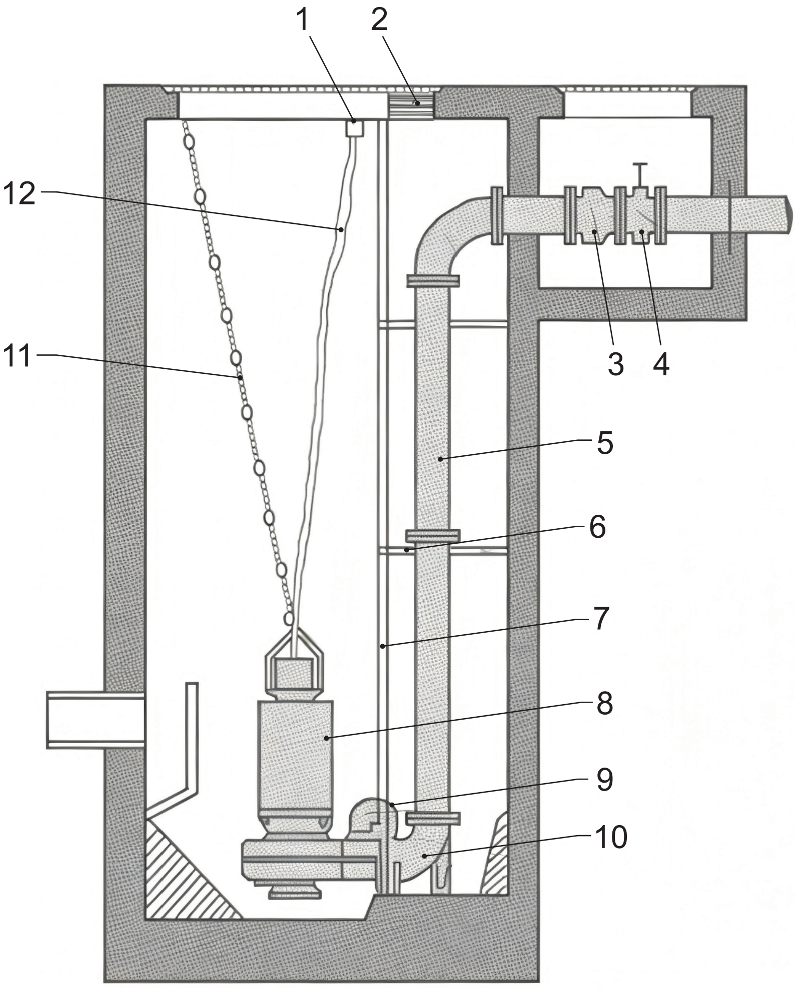
The discharge line must be installed in compliance with the relevant regulations. DIN 1986/10 and EN 12056 (or any applicable local regulations e.g. GB/T24674) applies in particular to the following:

- The discharge line should be fitted with a backwash loop (180° bend) located above the backwash level and should then flow by gravity into the collection line or sewer.
- The discharge line should not be connected to a down pipe.
- No other inflows or discharge lines should be connected to this discharge line.

	ATTENTION
	The discharge line should be installed so that it is not affected by frost.

9.2. Installation type

9.2.1. Concrete pump pit



1	Cable conduit
2	Upper holder
3	Check valve
4	Valve
5	Discharge pipe
6	Guide rail bracket
7	Guide rail
8	Grey water pump
9	Bracket
10	Pedestal
11	Chain
12	Cable

10. Electrical connection

CAUTION

All safety hints in other sections must be observed!

DANGER

Dangerous voltage

Before commissioning, an expert should check that one of the necessary electrical protective devices is available. Earthing, neutral, earth leakage circuit breakers etc. must comply with the regulations of the local electricity supply authority and a qualified person should check that these are in perfect order.

ATTENTION

The power supply system on site must comply with the local regulations with regard to cross-sectional area and maximum voltage drop. the voltage stated on the nameplate of the pump must correspond to that of the mains.

Suitably rated means of disconnection shall be incorporated in the fixed wiring by the installer for all pumps in accordance with applicable local national codes.

The power supply cable must be protected by an adequately dimensioned slow-blow fuse corresponding to the rated power of the pump.

DANGER

Dangerous voltage

The incoming power supply as well as the connection of the pump itself to the terminals on the control panel must comply with the circuit diagram of the control panel as well as the motor connections diagrams and must be carried out by a qualified person.

WARNING

The control panel shall only be operated when equipped with a motor overload protection relay and properly connected to the motor temperature protection switch or sensor.

Submersible pumps used outdoors must be fitted with a power cable of at least 10 meter length. Other regulations may apply in different countries.

In all installations, the power supply to the pump must be via a residual current device (e.g. RCD, ELCB, RCBO etc.) with a rated residual operating current in accordance with local regulations. For installation not having a fixed residual current device the pump must be plugged into the power supply through a portable version of device and not exceeding 30 mA.

All three phase pumps must be installed with motor starting and overload protective devices in the fixed wiring by the installer. Such motor control and protective devices must comply with the requirements of IEC standard 60947-4-1. They must be rated for the motor that they control, and wired and set/adjusted according to the instructions provided by the manufacturer. In addition, the overload protective device that is responsive to the motor current shall be set / adjusted to 125% of the marked rated current.

DANGER

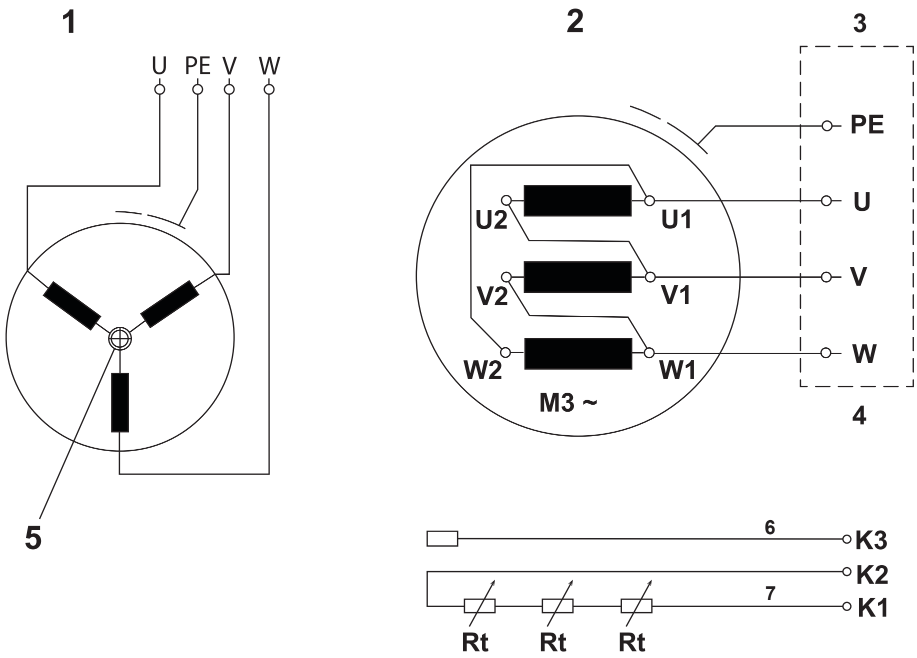
Dangerous voltage

Risk of electrical shock. Do not remove cord and strain relief and do not connect conduit to pump.

!	NOTE
	Please consult your electrician.

10.1. Wiring diagrams

Figure 2. Power and control cable wiring diagram



Legend

1 ≤4 kW motor

2 5.5 ~ 22 kW motor

3 Power cable

4 Control cable

5 Built-in thermal protector

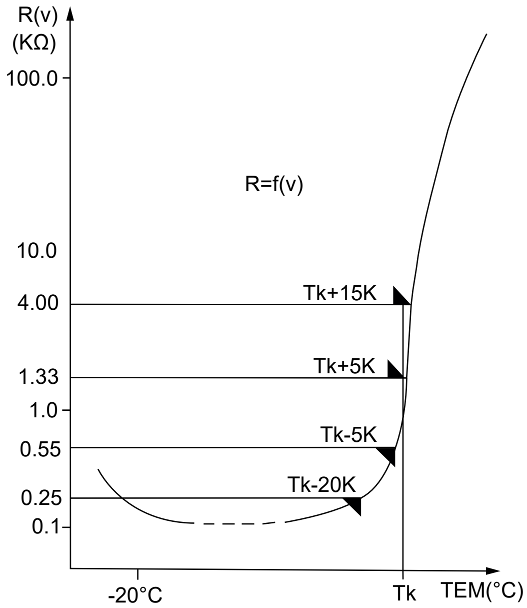
6 Leakage sensor

7 Winding PTC

Table 3. Control signals description

Control lead	Sensor	Function	Reference resistance	Reference trip resistance
K1 / K2	PTC	Winding temperature	R≤750 Ω	R≥3990 Ω
K3	Probe	Seal chamber moisture	R≥50 MΩ	Humidity≥95%, R≤33 KΩ

Figure 3. Curve of the working principle of thermistor (PTC)



	NOTE
!	PTC relays fitted in control panels and PTC sensors must be in accordance with DIN 44082.

	NOTE
!	Running the pump with the thermal and/or leakage sensors disconnected will invalidate related warranty claims.

11. Commissioning

	⚠ CAUTION
⚠	All safety hints in other sections must be observed!

Before commissioning, the pump should be checked and a functional test carried out. Particular attention should be paid to the following:

- Have the electrical connections been carried out in accordance with regulations?
- Have the thermal sensors been connected?
- Is the seal monitoring device correctly installed?
- Is the motor overload switch correctly set?
- Have the power and control circuit cables been correctly fitted?
- Is the pump pit cleaned out?
- Have the inflow and outflows of the pump station been cleaned and checked?
- Is the direction of rotation correct - even if run via an emergency generator?
- Are the level control switches functioning correctly?
- Are the required gate valves (where fitted) open?
- Do the non-return valves (where fitted) function easily?

11.1. Direction of rotation

11.1.1. Checking direction of rotation

When three phase units are being commissioned for the first time, and also when used on a new site, the direction of rotation must be carefully checked by a qualified person.

!	NOTE
	The start reaction is anti-clockwise.

!	ATTENTION
	When viewed from above, the direction of rotation is correct if the impeller rotates in a anti - clockwise manner.

⚠	⚠ CAUTION
	The direction of rotation should only be altered by a qualified person. When checking the direction of rotation, the pump should be secured in such a manner that no danger to personnel is caused by the rotating impeller or by the resulting air flow. Do not place your hand into the hydraulic system!

⚠	⚠ CAUTION
	When checking the direction of rotation, or when starting the unit, pay attention to the START REACTION. This can be very powerful and cause the pump to jerk in the opposite direction to the direction of rotation.

!	ATTENTION
	If a number of pumps are connected to a single control panel then each unit must be individually checked.

!	ATTENTION
	The mains supply to the control panel should have a clockwise rotation. If the leads are connected in accordance with the circuit diagram and lead designations, the direction of rotation will be correct.

11.1.2. Changing direction of rotation

⚠	⚠ CAUTION
	All safety hints in other sections must be observed!

⚠	⚠ CAUTION
	The direction of rotation should only be altered by a qualified person. If the direction of rotation is incorrect, alter it by changing over two phases of the power supply cable in the control panel. The direction of rotation should then be rechecked.

	WARNING
	Incorrect rotation direction leads to improper pump operation and cause equipment damage. Verify the correct rotation direction using the rotation detection device when operating with the main power supply or backup generator.

12. Maintenance and service

	CAUTION
	All safety hints in other sections must be observed!

	DANGER
	Dangerous voltage Before commencing any maintenance work the unit should be completely disconnected from the mains by a qualified person and care should be taken that it cannot be inadvertently switched back on.

	DANGER
	If the power cable Is damaged it must be replaced by the manufacturer, its service agent or a qualified person.

	CAUTION
	When carrying out any on-site service or maintenance work i.e. cleaning, venting, fluid inspection or changing, and adjustment of the bottom plate gap, the safety regulations covering work in enclosed areas of installations as well as good general technical practices should be followed.

	CAUTION
	Repair work must only be carried out by qualified personnel approved by Sulzer.

12.1. General maintenance instructions

Sulzer submersible pumps are reliable quality products, each being subjected to careful final inspection. Lubricated-for-life ball bearings, together with monitoring devices, ensure optimum pump reliability provided that the pump has been connected and operated in accordance with the operating instructions. However, should a malfunction occur, do not improvise, but ask your Sulzer Customer Service Department for assistance. This applies particularly if the pump is continually switched off by the current overload in the control panel, by the thermal sensors of the thermo-control system, or by the leakage sensor (DI).

Regular inspection and care is recommended to ensure a long service life. Service intervals vary for Sulzer units depending on installation and application. For recommended service interval details contact your local Sulzer Service Center. A maintenance contract with our Service Department will guarantee the best technical service.

When carrying out repairs, only original spare parts supplied by the manufacturer should be used. Sulzer warranty conditions are only valid provided repair work has been carried out in a Sulzer approved workshop, and original Sulzer spare parts have been used.

12.1.1. Inspection intervals

The pump must be inspected at regular intervals to ensure reliable operation. It is recommended that a functional inspection be carried out on a monthly basis.

Maintenance and servicing of the hoist must be performed by qualified personnel at the following intervals:

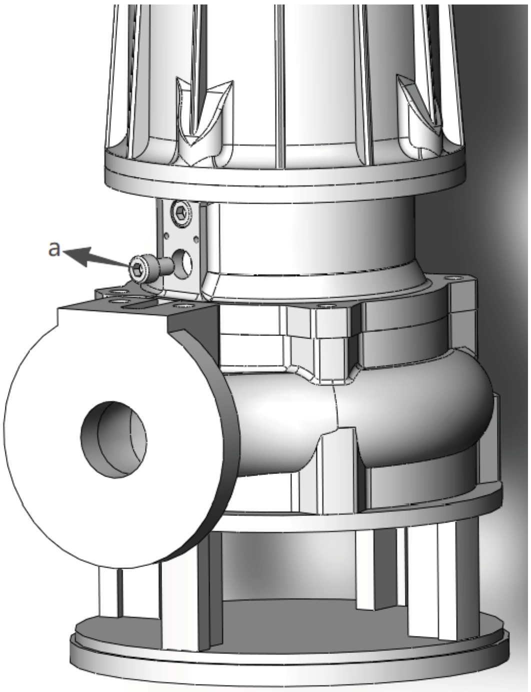
- Commercial facilities: every 3 months
- Apartment units: every 6 months
- Single-family housing: once every 12 months

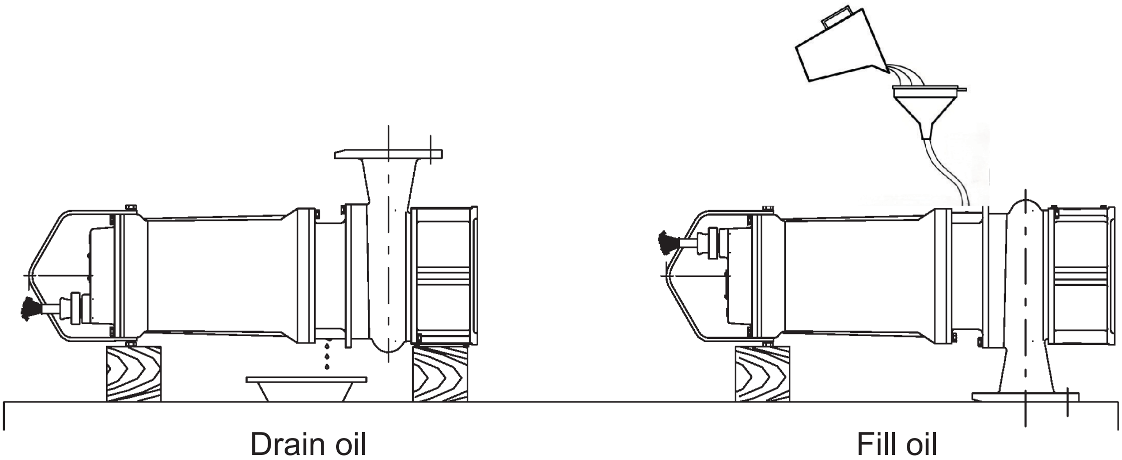
12.2. Drain and fill the lubricating oil

About this task

The seal chamber between the motor and the hydraulic section has been filled at manufacture with lubricating oil. An oil change is only necessary if a fault occurs.

Oil: White oil No. 7 VG7





Procedure

	WARNING
	Ensure that the motor is cold before draining and filling the lubricating oil.

1. Loosen the drain plug screw (a) enough to release any pressure that may have built-up, and re-tighten.

	CAUTION
	Before doing so, place a cloth over the plug screw to contain any possible spray of oil as the pump de-pressurizes.

2. Secure a hoist to the lifting point. Carefully lay the pump on its side and rotate it until the drain opening is positioned at the lowest point.

Note: If there is insufficient space to place a container beneath the drain opening,the oil must be drained into a sump.

3. Remove the plug screw and seal ring (a) from the drain hole.
4. After the oil is fully drained, place the pump in a horizontal position sitting on its discharge flange ensure the motor housing is properly supported from underneath.

	CAUTION
	To prevent the pump from toppling over ensure it is supported to lie flat on its discharge flange.

5. Select the required volume of oil from the quantities table and slowly pour into the drain hole.
6. Refit the plug screw and seal ring.

12.3. Oil quantities (liters)

Table 4.

Pump Model	Pump designation	Oil injection volume (L)
GWP/GWPS	0501-0502	0.3
	0503-0506	0.2
	0507-0508	0.4
	0509-0511	0.6
	0601-0602	0.4
	0603-0605	0.6
	0801-0805	0.2
	0806-0809	0.6
	0810-0816	1.0
	1001-1002	0.2
	1003-1005	0.4
	1006-1007	0.6
	1008-1012	1.0
	1013-1018	1.4
	1501	0.6
	1502-1504	1.0
	1505-1509	1.4
	2001	1.0
	2002-2005	1.4
	2501-2504	
GWPC	0501-0502	0.2
	0503-0505	0.4
	1001	
	1002-1003	0.6

12.4. Cleaning

If the pump is used for transportable applications, then in order to avoid deposits of dirt and encrustation it should be cleaned after each usage by pumping clear water. In the case of fixed installation, we recommend that the functioning of the automatic level control system be checked regularly. By switching the selection switch (switch setting “HAND”) the sump will be emptied. If deposits of dirt are visible on the floats then these should be cleaned. After cleaning, the pump should be rinsed out with clear water and a number of automatic pumping cycles carried out.

13. Troubleshooting guide

Table 5.

Fault	Cause	Fix
Pump does not run	Leakage sensor shutdown	Check for loose or damaged oil plug, or locate and replace faulty mechanical seal / damaged o-rings. Change oil.[1]
	Air lock in volute	Shake or raise and lower the pump repeatedly until resulting air bubbles no longer appear at surface level.
	Level control override	Check for float switch that is faulty or tangled and held in OFF position in sump.
	Impeller jammed.	Inspect and remove jammed object. Check gap between impeller and bottom plate and adjust if necessary.
	Gate valve closed, non-return valve blocked.	Open gate valve, clean blockage from non-return valve.
Pump switching on/off intermittently	Temperature sensor shutdown.	Motor will restart automatically when pump cools down. Check thermal relay settings in control panel. Check for impeller blockage. If none of above, a service inspection is required.[1]
Low head or flow	Wrong direction of rotation.	Change rotation by interchanging two phases of the power supply cable.
	Gap too wide between impeller and bottom plate	Reduce gap.
	Gate valve partially open.	Open valve fully.
Excessive noise or vibration	Defective bearing.	Replace bearing.[1]
	Clogged impeller.	Clear the pump blockage to remove and clean hydraulics.
	Wrong direction of rotation.	Change rotation by interchanging two phases of the power supply cable.
[1] Pump must be taken to approved workshop.		

CAUTION

Before commencing any inspection or repair work the pump should be completely disconnected from the mains by a qualified person and care should be taken that it cannot be inadvertently switched back on.

14. Company details

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